AWS Databases Lesson

Skills you will learn

Cloud Computing: Storage & Databases

Lesson Overview

In this lesson, you will explore the diverse database services offered by AWS, designed to cater to different application and data management requirements. We will cover services such as Amazon RDS, Aurora, DynamoDB, and ElastiCache, along with database migration tools like AWS DMS and AWS SCT.

Learning Objectives

By the end of this lesson, you will be able to:

1. Identify various AWS database services, including Amazon RDS, Aurora, DynamoDB, and ElastiCache.
2. Understand the capabilities and use cases of Amazon RDS.
3. Learn how AWS Database Migration Service (AWS DMS) and AWS Schema Conversion Tool (AWS SCT) facilitate cloud migrations.

1. AWS Database Services

AWS provides a wide range of purpose-built databases to meet different application needs. These include both **relational** and **non-relational** databases.

Relational Databases (SQL-based)

- **Amazon RDS** (Relational Database Service)
- **Amazon Aurora** (MySQL/PostgreSQL-compatible, enterprise-grade database)

Non-Relational Databases (NoSQL-based)

- **Amazon DynamoDB** (Key-Value and Document Database)
- **Amazon DocumentDB** (MongoDB-Compatible Database)

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- **Amazon Keyspaces** (Managed Apache Cassandra Database)
- **Amazon Timestream** (Time-Series Database)
- **Amazon Neptune** (Graph Database)

Caching & Data Acceleration

- **Amazon ElastiCache** (In-memory caching service supporting Memcached and Redis)
- **Amazon Redshift** (Managed Data Warehousing Service)

2. Amazon RDS: Managed Relational Database

What is Amazon RDS?

Amazon RDS (Relational Database Service) is a managed cloud-based service that simplifies database provisioning, scaling, and maintenance. It supports multiple database engines, including:

- **MySQL**
- **PostgreSQL**
- **MariaDB**
- **Oracle Database**
- **Microsoft SQL Server**
- **Amazon Aurora**

Key Features of Amazon RDS

✅ Fully managed by AWS (automatic patching, backups, and monitoring). ✅ High availability with Multi-AZ deployment. ✅ Encryption at rest and in transit. ✅ Automated scaling and performance optimization.

Example: Creating an RDS Database Instance

Copy Explain

```
aws rds create-db-instance \
  --db-instance-identifier mydbinstance \
  --db-instance-class db.t3.micro \
  --engine mysql \
  --allocated-storage 20 \
  --master-username admin \
  --master-user-password mypassword
```

3. Amazon Aurora: High-Performance Relational Database

Amazon Aurora is a fully managed **MySQL- and PostgreSQL-compatible** database that provides enhanced availability and performance.

Why Choose Aurora?

- **5x faster than MySQL, 3x faster than PostgreSQL.**
- **Replicates data across multiple Availability Zones automatically.**
- **Auto-scaling, with up to 15 read replicas.**
- **Serverless option available for on-demand scaling.**

Use Case:

- **Enterprise applications requiring high availability and durability.**
- **E-commerce websites handling high volumes of transactions.**

4. Amazon DynamoDB: Fully Managed NoSQL Database

Amazon DynamoDB is a key-value and document-based NoSQL database that is highly scalable and fully managed.

Key Features

✅ **Single-digit millisecond performance.** ✅ **Auto-scaling with on-demand and provisioned capacity modes.** ✅ **Global tables for multi-region replication.** ✅ **Built-in security and backup features.**

Example: Storing User Information in DynamoDB

Copy Explain

```
import boto3

dynamodb = boto3.resource('dynamodb')
table = dynamodb.Table('Users')

# Insert a new user
table.put_item(
   Item={
      'UserID': '123',
      'Name': 'Alice',
      'Email': 'alice@example.com'
   }
)
```

5. Amazon ElastiCache: In-Memory Data Caching

Amazon ElastiCache enables **fast, low-latency caching** for applications by supporting in-memory data stores like **Memcached** and **Redis**.

Use Cases

- Caching frequently accessed data to reduce database load.
- Real-time analytics and gaming leaderboards.
- Session storage for web applications.

Example: Creating an ElastiCache Cluster (Redis)

```
aws elasticache create-cache-cluster \
  --cache-cluster-id myrediscluster \
  --engine redis \
  --cache-node-type cache.t2.micro \
  --num-cache-nodes 1
```

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6. AWS Database Migration Services (DMS & SCT)

AWS provides tools to migrate databases to the cloud with minimal downtime.

AWS Database Migration Service (AWS DMS)

AWS DMS enables seamless migration of databases from on-premises or other cloud providers to AWS.

✅ **Supports homogenous (MySQL → MySQL) and heterogeneous (Oracle → PostgreSQL) migrations.**
✅ **Continuously replicates data during migration.**

Example: Creating a Database Migration Task

```
aws dms create-replication-task \
  --replication-task-identifier my-migration-task \
  --source-endpoint-arn arn:aws:dms:source-db \
  --target-endpoint-arn arn:aws:dms:target-db \
  --migration-type full-load \
  --table-mappings file://mappings.json
```

Copy Explain

AWS Schema Conversion Tool (AWS SCT)

AWS SCT automates schema conversion from one database engine to another. It is commonly used in **heterogeneous** migrations (e.g., Oracle to PostgreSQL).

✅ **Converts database schemas and SQL code automatically.** ✅ **Generates a migration assessment report.**

Conclusion

- **AWS offers both relational (SQL) and non-relational (NoSQL) databases to meet various workload requirements.**
- **Amazon RDS and Aurora provide managed relational database solutions with high availability.**
- **DynamoDB delivers a fast and scalable NoSQL option.**
- **ElastiCache improves application performance with in-memory caching.**
- **AWS DMS and AWS SCT simplify database migrations.**

🚀 **Well done! You've successfully explored AWS database services!** 🎉

AI TUTOR ACTIONS

Generate a practice quiz Summarize this unit Ask about this unit

Complete unit

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